AWS RDS Microsoft SQL Access Guide

Input Guide

This section outlines the input parameters required for provisioning AWS RDS Microsoft SQL using the DASH portal. Each parameter includes a description, available options, usage guidance, and notes on valid combinations when applicable.

Input Parameter	Description	Values Provided	Value Guidance	Valid Combinations
region	The region where the database instance will be deployed.	us-east-1, us-west-2, eu-west-1, ap-south-1, ca-central-1, eu-west-2, ap-southeast-1, sa-east-1.	Choose the region that is geographically closest to your end users or services to minimize latency and improve performance.	N/A
db_identifier	A unique name to identify the database instance.	A unique name across all DB instances owned by your account in the current AWS Region.	The DB instance identifier is case-insensitive, but is stored as all lowercase (as in "mydbinstance") **Constraints**: 1 to 63 alphanumeric characters or hyphens. The first character must be a letter. Can't contain two consecutive hyphens. Can't end with a hyphen.	N/A
db_username	The master username for database administration.	Alphanumeric string.	1 to 16 alphanumeric characters. The first character must be a letter.	N/A
db_password	The password	Alphanumeric	Set a strong	N/A

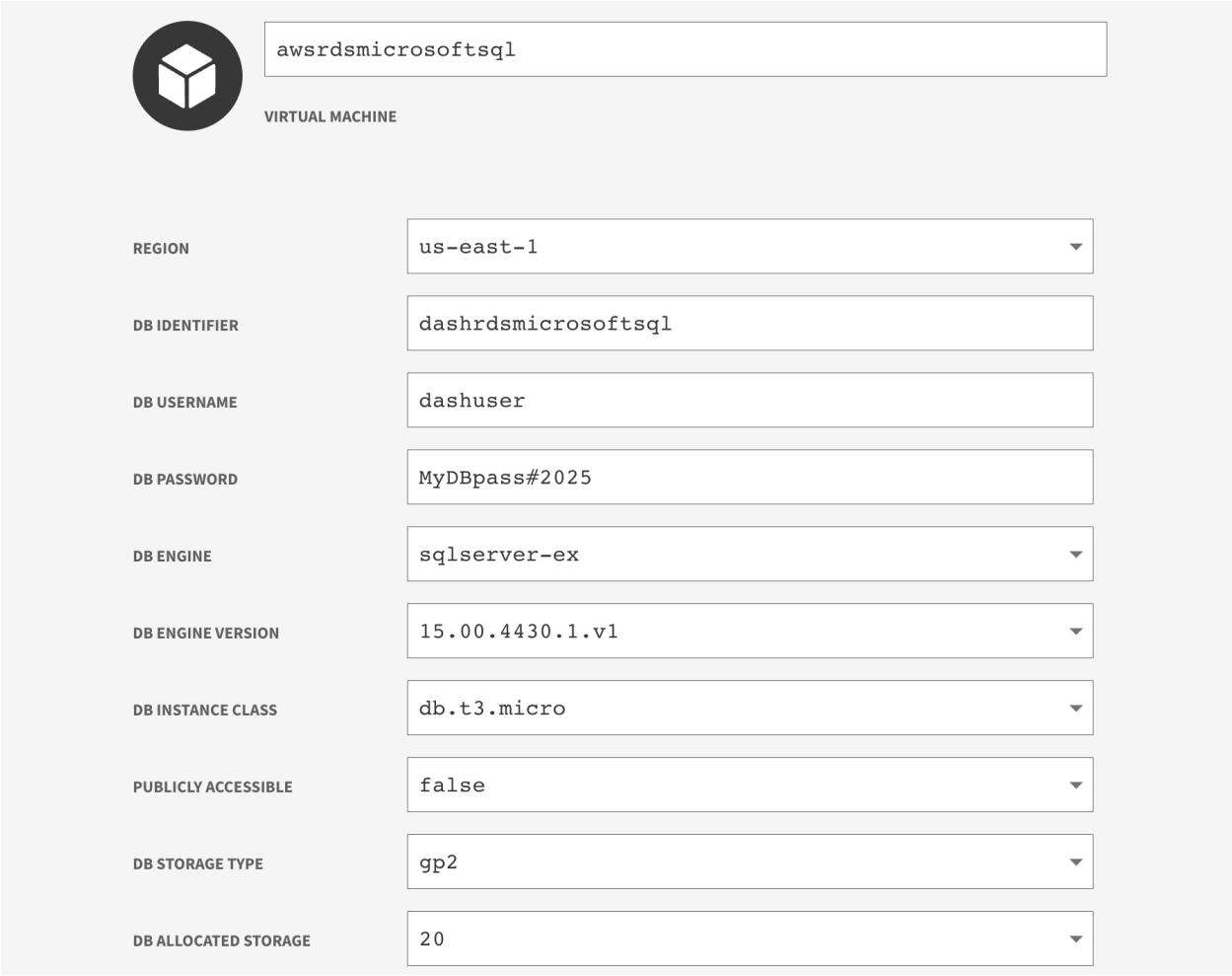
	associated with the master username for secure access.	string.	password for the master username. Minimum constraints: At least 8 printable ASCII characters. Can't contain any of the following symbols: / ' " @ . Example (valid): MyDBpass#2025	
db_engine	The database engine edition to use.	**Editions:** sqlserver-ex, sqlserver-web, sqlserver-se, sqlserver-ee	**Express Edition:** Affordable database management system that supports database sizes up to 10 GB. **Web Edition:** Supports public and Internet-accessible webpages, websites, web applications, and web services. **Standard Edition:** Core data management and business intelligence capabilities for mission-critical applications and mixed workloads. **Enterprise Edition:** Comprehensive high-end capabilities for mission-critical applications with demanding database workloads and	

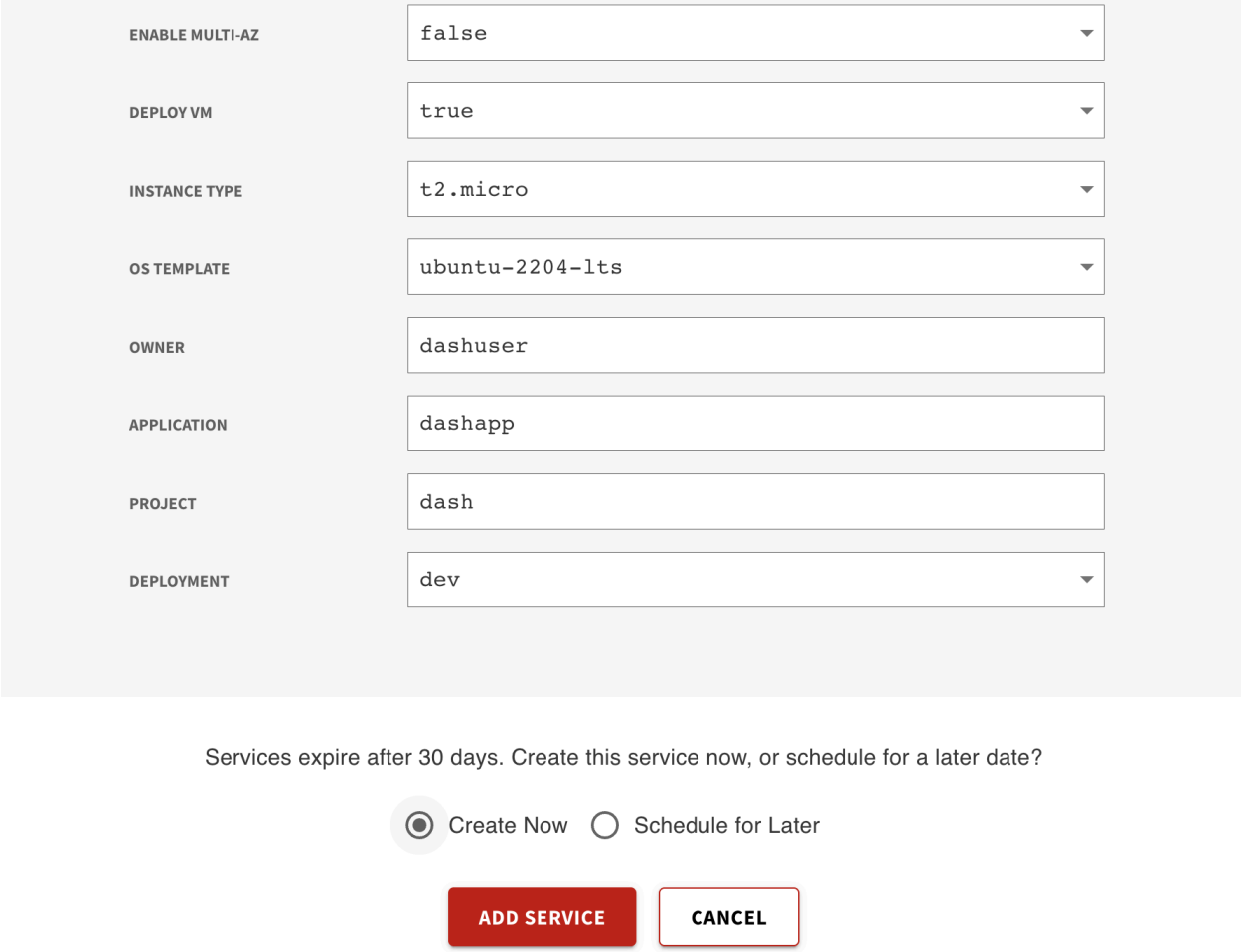
			business intelligence requirements.	
db_engine_version	The version of the Microsoft SQL database engine.	**Versions**: 15.00.4430.1.v1, 15.00.4420.2.v1, 15.00.4415.2.v1, 15.00.4410.1.v1, 15.00.4395.2.v1.	Choose based on Microsoft SQL requirements.	N/A
db_instance_class	The compute and memory capacity allocated for the database instance.	**Instance Configuration:** db.t3.micro, db.t3.medium, db.t3.large, db.m5.large, db.m5.xlarge, db.m5.2xlarge, db.m5d.large, db.m5d.xlarge, db.m6i.xlarge, db.m6i.2xlarge	Choose based on workload requirements. **Note:** sqlserver-ex supports only Burstable classes.	**Instance Configuration supported by sqlserver-express-edition:** db.t3.micro, db.t3.medium, db.t3.large.
publicly_accessible	Determines whether the database can be accessed from the public internet.	**Boolean:** True, False	Choose true if external access is needed, otherwise false.	N/A
db_storage_type	The type of storage allocated for the database.	**Storage types:** gp2 (or) gp3	Choose based on performance and cost requirements.	N/A
db_allocated_storage	The amount of storage (in GB) to be provisioned for the database.	**Allowed values (in GB):** 20, 40, 60, 80, 100.	Choose based on workload requirements.	N/A
enable_multi-az	Enables high availability by deploying the database in multiple Availability Zones.	**Boolean:** True, False.	Choose true for high availability or false for single-AZ.	N/A

deploy_vm	Whether an ec2 instance should also be provisioned to connect to the DB.	**Boolean:** True, False	Choose whether to set up a connection to a compute resource for this database.	N/A
instance_type	The size and configuration of the EC2 instance, which defines the amount of vCPUs, memory, and networking performance allocated.	t2.nano, t2.micro, t2.small, t2.medium, t2.large, t2.xlarge, t2.2xlarge, a1.metal, c4.large, c4.xlarge.	Choose based on workload requirements (vCPU, memory).	N/A
os_template	Pre-configured base operating system image (Amazon Machine Image) used for provisioning the EC2 instance.	ubuntu-2204-lts, ubuntu-2004-lts, ubuntu-2404-lts, rhel-9, rhel-8, centos, amazon-linux, amazon-linux-2023.	Choose an OS required for your workload.	N/A
owner	The team or individual responsible for maintaining the resource.	Specify the team name or individual responsible for maintaining the instance.	Provide correct team/individual name.	N/A
application	Associated service or application name for the resource.	Enter the associated service name linked to the instance.	Provide a service name linked to the instance.	N/A
project	Project or initiative name linked to the resource.	Indicate the project or initiative name related to the resource.	Provide the project name related to the resource.	N/A
deployment	The target environment (e.g., development,	Define the deployment environment name (e.g., dev,	Choose the environment where the database has to be deployed.	N/A

	testing, prod) where the DB will be provisioned.	test, prod)		
Services expire after 30 days. Create this service now, or schedule for a later date?	RDS Microsoft SQL will be deleted 30 days after creation, or it can be scheduled to be created at a later date.	Create Now (or) Schedule for Later	Choose based on RDS Microsoft SQL requirements.	N/A

Sample Input Selection:





Output Guide

This section outlines the output values generated after AWS RDS Microsoft SQL provisioning through the DASH Portal. These outputs can be used to access, manage, and connect to your Microsoft SQL Database.

Outputs	Description	Example Value
endpoint	The database endpoint used to connect to the DB instance.	dashrdsmicrosoftsql.czeq0oqa6rxx.us-east-1.rds.amazonaws.com
public_ip	The public IP address of the provisioned EC2(if enabled).	54.161.70.126
sqlcmd_connection	The Microsoft SQL connection string includes endpoint, port and username to connect to the DB.	sqlcmd -S tcp:dashrdsmicrosoftsql.czeq0oqa6rxx.us-east-1.rds.amazonaws.com,1433 -U

		dashuser
ssh_password	The password associated with the SSH username for EC2 access.	sqtEkxw3
ssh_username	The username to connect to the EC2 via SSH.	dash

Note: If **deploy_vm** is set to **false** in the input, the output will only include the endpoint and sqlcmd_connection.

What you can do with these outputs:

1. When publicly accessible is enabled:

You can connect directly to the DB endpoint.

sqlcmd -S tcp:<endpoint>,1433 -U <username>

Example: sqlcmd -S tcp:dashrdsmicrosoftsql.czeq0oqa6rxx.us-east-1.rds.amazonaws.com,1433 -U dashuser

When prompted, enter the value of db_password (which you provided in the input) to connect to the database.

2. When publicly accessible is disabled:

You must first connect to the EC2 and then access the DB from inside it.

Step 1: SSH into the EC2

Example: ssh <ssh_username>@<public_ip>

Replace <ssh_username> and <public_ip> with the values generated in the outputs. When prompted, enter the value of ssh_password to login to the EC2.

Step 2: Connect to the DB engine from the EC2

sqlcmd -S tcp:<endpoint>,1433 -U <username>

Example: sqlcmd -S tcp:dashrdsmicrosoftsql.czeq0oqa6rxx.us-east-1.rds.amazonaws.com,1433 -U dashuser

When prompted, enter the value of db_password (which you provided in the input) to connect to the database.